Chemical Engineering Journal 452 (2023) 139190

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Optical and experimental evaluation of a directly irradiated solar reactor for the catalytic dry reforming of methane

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ARTICLE INFO

Keywords:
Solar energy
Ray-tracing
Dry reforming methane
Experimental design
Concentrated light

ABSTRACT

Dry reforming of methane is a process to produce syngas which is a major precursor for many chemicals and ultra-clean fuels. It is a CO_2-assisted process that leads to the conversion of CO_2 to higher-value products but is also a highly endothermic process that requires significant amounts of energy in the form of CO_2 emitting fuels. For these reasons, this study focused on the direct utilization of concentrated solar energy by irradiating the catalyst directly rather than via a heat transfer fluid or conductive heat transfer. In addition, it uniquely adopts a tubular reactor with transparent (quartz) walls configuration, which allows extending the length of the irradiated (hot) zone to control the residence time. The new reactor was designed using Monte-Carlo ray-tracing modeling and evaluated experimentally using a commercial Ni-based catalyst. In brief, the reacting gas mixture (CH_4:CO_2 – 5 %: 5 %) was fed into the reactor at weight gas hourly space velocities of around 17–58 l h^{-1} g^{-1} and three different temperature levels (550 °C, 650 °C, and 800 °C). Although the non-conventional mode of heating, the achieved methane conversions (~93 % at 800 °C to ~ 53 % at 550 °C) and H_2 / CO ratios (~0.9 at 800 °C to ~ 0.4 at 550 °C) were similar to literature studies with the same catalyst. At the same time, the quartz walls showed no degradation after more than 80 h of intermittent testing. On the other hand, the overall energy efficiency was estimated to be less than 1 %, considering the radiation intercepted by the reactor system. However, it rose to 5 % – 25 % when correcting for the actual amount of irradiance on the catalytic bed.

1. Introduction

The influence of CO_2 emission on greenhouse and the increasing demand for fuel with the noticeable shortage of oil and gas motivate society to explore alternative energy resources [1]. Among the different alternatives, the syngas (H_2 and CO) has been proved to be promising [1] as a major precursor to producing ultra-clean fuels and many chemicals [2]. Methane reforming, e.g., steam reforming (SMR) and dry reforming reaction (DRR), were [3] and appear to remain [4] the industrially preferred technologies for producing both syngas and H_2. However, these reactions are highly endothermic and today the required energy is provided by the combustion of fossil fuels. Nonetheless, the process turns out to be achievable and economical with the development of new catalysts and the inclusion of renewable energy, e.g. solar [5]. Solar energy has been used in the methane reforming process indirectly via the conversion to electric power (photovoltaics) and directly as photonic and thermal power [6]. The essential advantage of solar energy is the replacement or reduction of fossil fuel consumption while simultaneously upgrading the reactants energy content by 22–28 % and reducing CO_2 emissions [7]. For these reasons, the field attracts significant interest, frequently reviewed in the international literature [8,9]. The three main challenges lie in the catalyst performance and stability, in the operational character of the process (continuous vs batch), and in the solar reactor itself [10,11]. The latter is one of the central aspects of this work.

The role of the solar reactor system is to efficiently “absorb” the irradiation (optical design aspects), uniformly heat the catalytic materials and reactants (heat transfer design aspects), and maximize the contact between the reacting gas phase and catalyst (mass transfer design aspects) [2]. These aspects can negatively affect the solar reformer’s performance and long-term operation stability [12]. Solar reformers (reactors) are classified to direct and indirect systems depending on whether the fluid or particles contact directly or indirectly the solar radiation [13]. Indirectly heated systems are those in which the solar receiver and reactor are two separate units or systems. The solar energy

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https://doi.org/10.1016/j.cej.2022.139190
Received 27 May 2022; Received in revised form 4 September 2022; Accepted 9 September 2022
Available online 14 September 2022

Nomenclature

English Letters

E#	Index for experiment # (see Table 2)
GHSV	Gas hourly space velocity [l hour^{-1} $g_{cat.}^{-1}$] or [hour^{-1}]
f_i	feed molar rate of species i [mol/s]
L#	Operational level # for experimental factor (see Table 1)
W_{chem}	power stored as chemical enthalpy [kW]
$W_{inc,ap}$	power of incident light on the reactor's aperture [kW]
$W_{inc,bed}$	power of incident light on the catalytic bed [kW]
x	extent of reaction
X	methane conversion to synthesis gas [-]
y_i	mole fraction of species i [-]
η_{chem}	light to chemical energy efficiency [-]
η'_{chem}	adjusted energy efficiency [-]

Greek Letters

ΔH	Enthalpy of reaction [kJ/mol]

is used to heat some intermediary heat transfer fluid and then the reactor [2]. The directly heated systems where the solar receiver and reactor are one integrated unit, and thus the reactor is irradiated by the solar energy. These can again be classified into full-direct and direct–indirect heated systems if the catalytic or reacting materials are irradiated or just the reactor walls, respectively [13,14]. The former type of reactors has extra benefits, such as higher heat transfer rates but different challenges due to the requirement for transparent walls [15]. The reported reactor configurations and solar-to-chemical energy conversion vary. Chuayboon et al. [16] reached a greater than 5 % efficiency with a directly-irradiated volumetric solar CeO absorber at around 1000 °C and stressed the significant heat losses. Later, the same group replaced CeO with ZnO and showed the potential of coupling methane reforming to syngas with Zn production while maintaining the same solar-to-chemical energy efficiency [17]. On the other hand, the indirect or direct–indirect systems promote hybrid solar and electric energy heated reactors to allow for an uninterrupted operation. In Ma et al. [18], sodium served as the heat transfer medium in a wick structure achieving a 100 % steam reforming methane conversion and 45 % solar to thermochemical energy efficiency. An additional claim highlighted the remarkably short start-up time when a 458 kW m^{-2} irradiance was applied compared to traditional indirect systems. While most of the above systems operate at elevated temperatures (greater than500 °C), it appears possible to achieve efficiencies of 30 % at lower temperatures (425 °C) by integrating a membrane reactor with a parabolic trough solar collector [19]. Nevertheless, venous solar reactor designs and configurations have been investigated and optimized recently, mainly with numerical tools, for CO_2 reduction with methane. Examples are the biomimetic venous [20,21], biomimetic leaf-type [22], and foam [23–25] hierarchical porous structure reactors.

In parallel to the efficient heating of the reforming catalyst system, the direct irradiation opened the opportunity to exploit the photonic part of the solar energy. However, the indications in earlier studies were inconclusive, if not against it [7,26], probably because of the lack of photoactive components. [27] illustrated the synergistic photothermal dry reforming of methane at different irradiance levels (zero to ~ 31 suns or 31 kW m^{-2}) and temperatures with a Ni/TiO_2 catalyst without providing estimations for the energy efficiency of the system. Instead, using new photoactive materials, high light-to-fuel efficiencies for CO_2 reduction were reported for Ni nanocrystals modified with SiO_2-clusters [28], nanocomposite of Co on Co-doped Al_2O_3 nanosheets [29], Ni/Co alloy also on Co-doped Al_2O_3 nanosheets [30], and Ni nanoparticles on Mg-doped Al_2O_3 [31]. Important to realize that the photo-thermal effect on the kinetic and mechanistic aspects increases the complexity of these studies, given the breadth and diversity of the pure thermal energy reaction mechanisms and kinetics, as reviewed by Kathiraser et al. [32]. Especially when the roles of photoactivation and, in general, light-driven thermocatalysis are still under investigation Wu et al. [33]. A challenge that was also identified in the recent review of Li et al. [34] on the coupled thermo-photo methane reforming, in addition to the challenges of the small-scale laboratory systems and the low reactant conversions. Indeed, optimizing the process parameters at industrial scale level is critical to couple simulations with accurate kinetic models; for example, Chen and Gan [35] on the methane reforming under partial oxidation conditions.

Different types of catalysts have been tested for DRR. Some examples are metal catalysts supported by ceria oxides or ceria itself because of its excellent oxygen storage capability [36–38]. Other metal oxides have also been suggested as catalysts and supports, such as ZnO, ZrO_2, CeO_2, FeO, and Al_2O_3 [39]. Among them, Ni-based catalysts on different types of supports are some of the most common catalytic systems, despite their deactivation and coke formation [40]. Specifically for the DRR mechanism on Ni- catalysts, the formation of coking appears to be a sensitive function of the shape and reconstruction of the catalyst particle during the reaction as well as steps and kinks on the exposed Ni surface [41]. Therefore, it is vital to investigate the effects of direct concentrated light on the catalyst particle and the whole system's performance. Recently, Zheng et al. [42] studied mathematically a directly irradiated porous catalytic absorber (cylindrical cavity) to investigate and optimize the reactor design and irradiance delivery characteristics and concluded that the non-uniform distribution of irradiance limited the methane conversion to a max of 71 %.

For these reasons, we propose a unique tubular reactor configuration with transparent walls, which hypothetically should allow for the full-direct and uniform heating of any length reaction bed. As described earlier, most reported solar reactor designs with similar features have not been evaluated experimentally or resolved to indirect irradiation; for example, by using a heat-transfer fluid or irradiating the reactor walls. Therefore, this work investigates and discusses preliminary experimental and numerical findings of an in-house developed and fabricated ~ 3 kW reactor, driven by direct irradiation from a high flux solar simulator (HFSS), for the dry reforming of methane. At first, in the methodology section, the optical modeling and design of the full-directly irradiated solar reactor are presented, including details of the testing and characterization methods. Then, in the results section, the syngas production and quality assessment at different temperature levels and gas hourly space velocities (GHSV: 865, 1270, and 2900 h^{-1}) are described for the DRR of a gas mixture (5 % CH_4, 5 % CO_2, 2 % Ar and He balance) passing through the catalytic fix bed (20 % Ni on Al_2O_3 support). Finally, in the discussion section, the implications of the concentrating light pattern on the system's energy efficiency and potentially the catalyst are discussed along with approaches to improve the performance.

2. Methods and materials

In-vitro solar thermochemical experiments utilize one or more artificial light sources with reflector(s) to concentrate the light on the reactor and provide radiant energy. The former parts (light source and reflector) are referred to as high flux solar simulator (HFSS), and the latter (reactor) as receiver of the concentrated light. The various aspects and concepts of solar thermochemical reactors [43] and HFSS [44] have already been reviewed. Thus, this chapter details the radiant energy source (i.e., HFSS) and the optical model developed to estimate the irradiance. The optical model was also utilized in the design of the solar thermochemical reactor, which is presented in the second section along with its instrumentation. The third section outlines the irradiance measurement procedure and equipment, one of the most important experimental aspects of this work. The final two sections elaborate on the experimental design and the selection of the efficiency metrics, and the catalyst materials and their characterization.

2.1. Radiant energy and optical modeling

The radiant energy was provided to the reactor by a high flux solar simulator (HFSS). It consists of seven independent lamp blocks with a truncated ellipsoidal and aluminum coated reflector (Fig. 1A), a 6 kW Xe short-arc, and an ozone-free lamp (XBO 6000 W/HS XL OFR by Osram), and the necessary power electronics and controller. The seven lamp blocks are placed on a "squeezed" hexagon configuration with the seventh block placed in the center (Fig. 2**A**). The HFSS is enclosed in a 7x3.5x2.5 m^3 (LxWxH) container with prefabricated aluminum composite panels for increased user safety, reduced light bleeding, and temperature control (Fig. 2**A** inset). Despite their different orientation, all lamp blocks are the same. The ellipsoidal reflectors (Fig. 1**A**) have a focal point at 90 mm and another at 2,970 mm from the vertex on the major (also horizontal) axis, 450 mm length, and a 735 mm opening. When the lamp is placed at the first focal point and the reactor, or any other target object, at the second focal point, the HFSS yields maximum irradiance. However, the maximum and optimum power distributions on the reactor depend on the reactor's geometry and materials' optical properties.

Therefore, the optical design, analysis, and optimization of the system were conducted using software (TracePro 2018 by Lambda Research Corporation) employing the Monte-Carlo ray tracing method. The objective of the optimization was to maximize the optical efficiency (ratio of the incident over emitted power) and the uniformity (reciprocal of irradiance's standard deviation). In the literature, there is a significant number of such software, both open-source and commercial. For example, SolTrace is an open-source software developed by the National Renewable Energy Laboratory (NREL) to model concentrating solar power systems particularly those with complex geometries using Monte Carlo ray tracing. CUtrace is another Monte Carlo parallel ray tracer code available on MATLAB Central used for radiation analysis and view factor determination especially for solar simulators [45]. Similarly, other software available includes Tonatiuh, OptiCAD, STRAL, and SPRAY, reviewed earlier by Osório et al. [46]. TracePro was selected because of the user-friendly environment, batch simulations, and availability in our Lab. Interested readers can find out the details of Monte-Carlo ray tracing modeling in any of the literature mentioned above.

Each of the seven lamp blocks (Fig. 1**A**) was modeled to obtain the intensity and distribution of the incident radiation flux on the reactor. The representation of the lamp's arc (source of light) is a challenge for HFSS modeling, with various approximations being suggested and tested e.g., spherical, cylindrical [48], and combinations [49]. Here, we employed a detailed arc representation using a combination of curved surfaces that mimic the brilliance distribution provided by the manufacturer [47] with excellent agreement with the experimental flux maps according to earlier studies [50,51]. In particular, the detailed but qualitative schematics by the lamp manufacturer (Fig. 1**B**) show that the first four inner layers have the highest brilliance (from inside to outside: 200, 150, 100 & 75 $kcd.cm^{-2}$) and emit, together with the plasma ball, approximately 94 % of the light produced in the arc. According to the same manual, the plasma ball alone accounts for 50 % of the light produced in the arc. Although the lamp manufacturer does not provide the specific dimensions of the plasma ball and the brilliance layers, these were estimated using the anode and cathode dimensions for the specific lamp model. The respective power output from each layer was introduced in the optical model by distributing the total power (6 kW) based on the layer's area and brilliance.

The lamp sits on a three-axis slider which allows for small movements (~0.06 mm) and, conversely, manipulation of the achieved irradiance levels and pattern. Recently, we presented a novel computational approach to control irradiance [52]. Unfortunately, that approach was unavailable during this experimental campaign, and the HFSS was fine-tuned manually.

2.2. Reactor configuration and instrumentation

The designs of the reflector, light source, and reactor employed herein are illustrated in Fig. 1. The former designs are standard, whereas the reactor was designed (Fig. 1**C**) exclusively for this study and to investigate the central idea. Specifically, it was based on the fixed bed tubular configuration with the appropriate additions and modifications

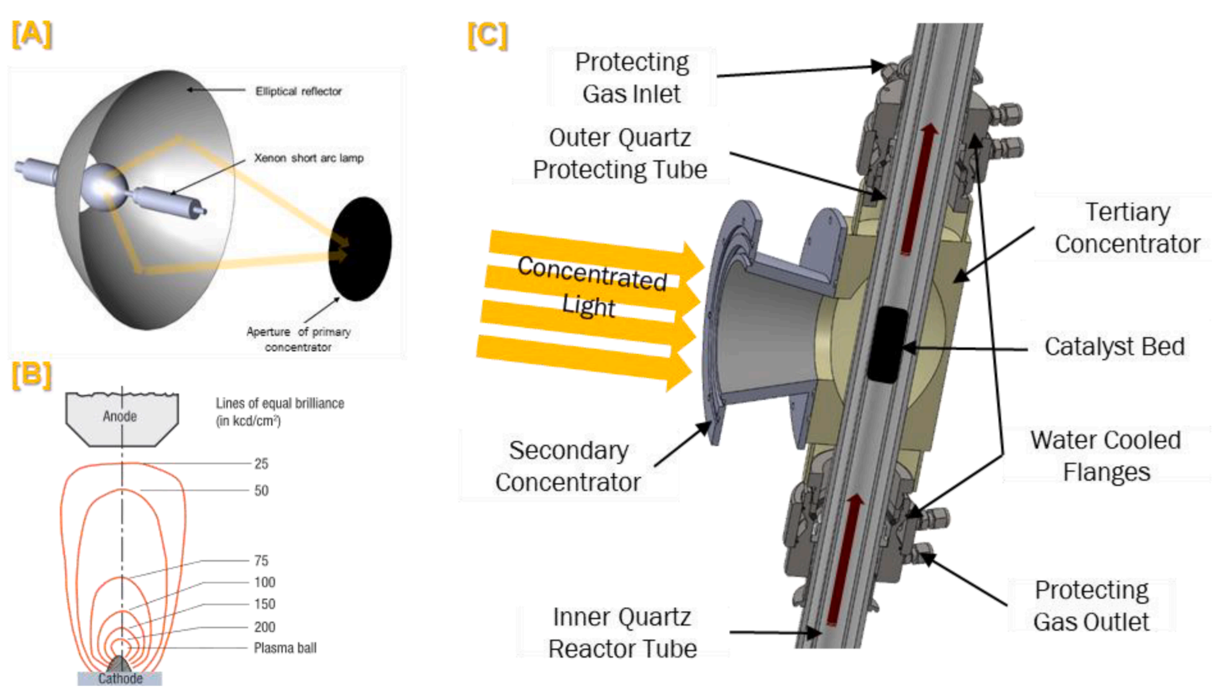

Fig. 1. Illustrations A) of the ellipsoidal reflector, B) of the brilliance distribution from a XOB® Xe-arc lamp (reconstructed from the manufacturer's manual [47]), and C) cross-section of the assembled reactor configuration (the water flanges for the inner tube are omitted).

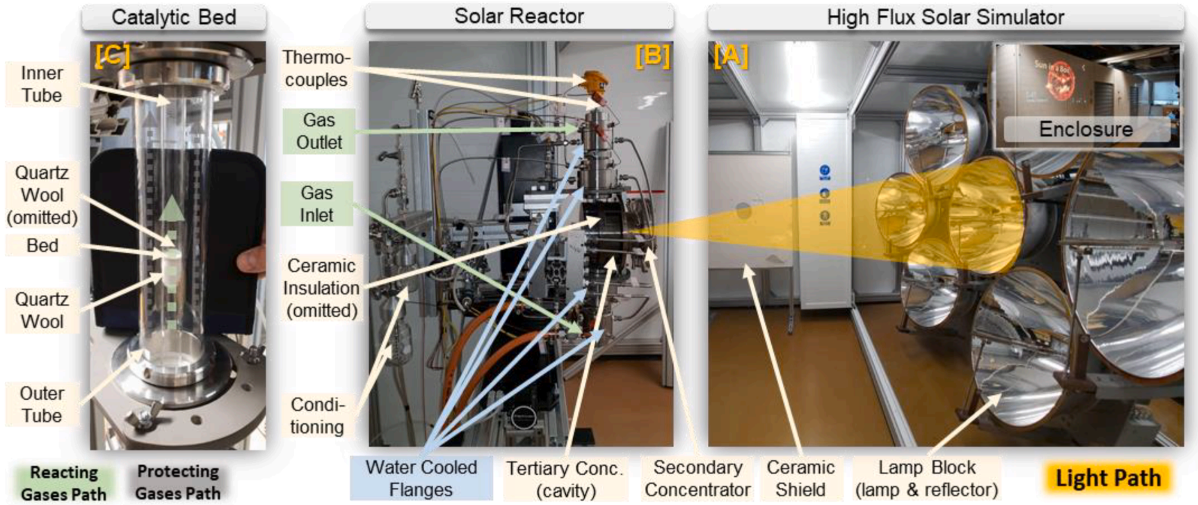

Fig. 2. Photos A) of the high flux solar simulator, B) of the assembled reactor setup (without the insulation), and C) of the reactor tubes and catalytic bed.

to allow direct and uniform irradiation of the catalyst bed. At first, the reacting zone's diameter (16 mm) and height (20 mm) were arbitrarily selected following available hardware and literature examples. After a few preliminary reactor designs and simulations with the optical model, two more light concentrators were added to improve the uniformity of the irradiance. Summing up to three concentrators including the lamp reflector, i.e., a primary (reflector), secondary, and tertiary concentrators. The secondary concentrator had a conical shape and was made of chromium-plated stainless steel (SS304), also used in an earlier study [50]. The tertiary concentrator was new, made of polished steel, and comprised two pieces forming a compound parabolic cavity with three openings to allow for the illumination of the catalyst bed and passage of the tubular parts hosting the bed (Fig. 1**C**). The final dimensions of the tertiary concentrator were selected after multiple simulations with the optical model considering the maximization of the incident power, the irradiance uniformity, and the manufacturability with Computer Numerical Controlled (CNC) machining. The outer walls of the tertiary concentrator were lined with firebrick material (~20 mm thickness, ~0.2 W m^{-1} K^{-1} @ 816 °C, purchased from McMaster-Carr).

The tubular part of the reactor was made of a quartz tube (16 mm ID, 1.5 mm wall thickness, 1,000 mm length by Robson Scientific UK) and hosted the catalyst bed. The ends of the tube were sealed perimetrically with two custom-made water-cooled flanges (SS316, Fig. 2**B**), each with four ports for gas pipe connections and thermocouples. The side of the flanges, opposite the tube sealing end, had an adaptor to permit thermocouples to reach the catalyst bed. An outer quartz tube (OD 56 mm, 2.0 mm thickness) was installed, encompassing the inner tube without blocking the illumination of the bed (Fig. 1**C** and Fig. 2**B**) and maintaining an inert (nitrogen) gas atmosphere outside the inner tube for safety reasons. The two ends of the outer quartz tube were also sealed with custom water-cooled flanges (SS316) each with four ports for gas pipe connections and thermocouples. These flanges were connected with the inner tube's flanges with a tri-clamp and a PTFE gasket for sealing. Fig. 2**B** offers a side-view photo of the fully assembled solar reactor. A ceramic fiber board (ST-1500 by Shine Technology Co. ltd, China) was placed as a radiation shield (Fig. 2**A**) between the reactor and the lamp blocks with an appropriate opening to allow irradiation of the reactor while protecting the adjacent components.

Except for the catalyst loading and unloading, the operation, monitoring, and control of the overall experimental system were carried out over two separate but interlocked LabView (by National Instruments) environments, one for the HFSS system and another for the reactor system, each on a separate personal computer. A description of the former environment and the complex HFSS system is available at Haseler et al. [52] Haseler et al. [51]. The system consisted of two main sub-systems, one for the temperature data and another for the flow and pressure data. Fig. 3 presents the overall process and instrumentation diagram (PID) of the reactor system.

In particular, the catalyst bed temperature was measured with two type-S (OD 6 mm, IEC60584 Class 1: ~± 1.0 °C, by Analysis ltd Greece) and long thermocouples (TI-5 and TI-6 in Fig. 3) with ceramic insulation. The bottom thermocouple supported the whole bed, and the top thermocouple was immersed in the catalyst bed. The temperatures of the reacting and protective gases (inlets, outlets, and after conditioning) were monitored using five type-N thermocouples (TI-1, 2, 3, 4, and 10) with Nicrotherm™ sheath (OD 1.5 mm, ICE60584 Class 2: ~±2.5 °C, by TCDirect UK). Each thermocouple was introduced directly into the gas flow through the dedicated ports on the flanges. The water outlet on each side of the flanges was checked using two type-K thermocouples (TI-11 and 13) with Inconel™ 600 sheath (OD 3 mm, IEC60584 Class 1: ~± 1.0 °C by Omega Engineering Inc UK). The latter thermocouples were selected based on in-house availability rather than their material and type, different from the rest of the thermocouples. In addition, two more flat type-K thermocouples were placed directly in contact with the peripheral surfaces of the secondary and tertiary concentrator and one more on the back surface of the tertiary concentrator. The voltage output of all the thermocouples was measured, converted to temperature (analog to digital error less than 0.25 °C), and collected with a 1 Hz sampling rate using a temperature input module (NI 9213 by National Instruments).

The gases' mass flow rates were regulated by four thermal mass flow controllers (type GFC 17 by AALBORG Instruments & Controls, Inc US) in the range of 0–5 standard l min^{-1} (0.02 l min^{-1} precision, accuracy 1.0 % of the full scale) and calibrated for the specific gases (N_2, He, CO_2). For the reacting gas mixture (CH_4: CO_2: Ar of 5 %: 5 %: 2 % and He balance), the same mass flow controller was used but calibrated in-house against a bubble flowmeter within the same range and achieving the same accuracy. Nevertheless, this introduced a slight displacement between the flow controller set points and the actual flow rates (i.e.,0.58 l min^{-1} instead of 0.6 l min^{-1}, 0.85 l min^{-1} instead of 0.9 l min^{-1}, 1.94 l min^{-1} instead of 2.0 l min^{-1}). The differential and back pressures of either tube (inner and outer) were measured using two pairs of relative (0–1 bar; PR33X) and differential (±200 mbar; PD33X) pressure transducers (accuracy 0.1 % of the full scale, by Keller and purchased from Comer srl, Italy). The analog signals from the flow controllers and pressure sensors transducers were measured and collected with a 1 Hz sampling rate using an analog input module (ADAM-4117 by Advantech, Greece).

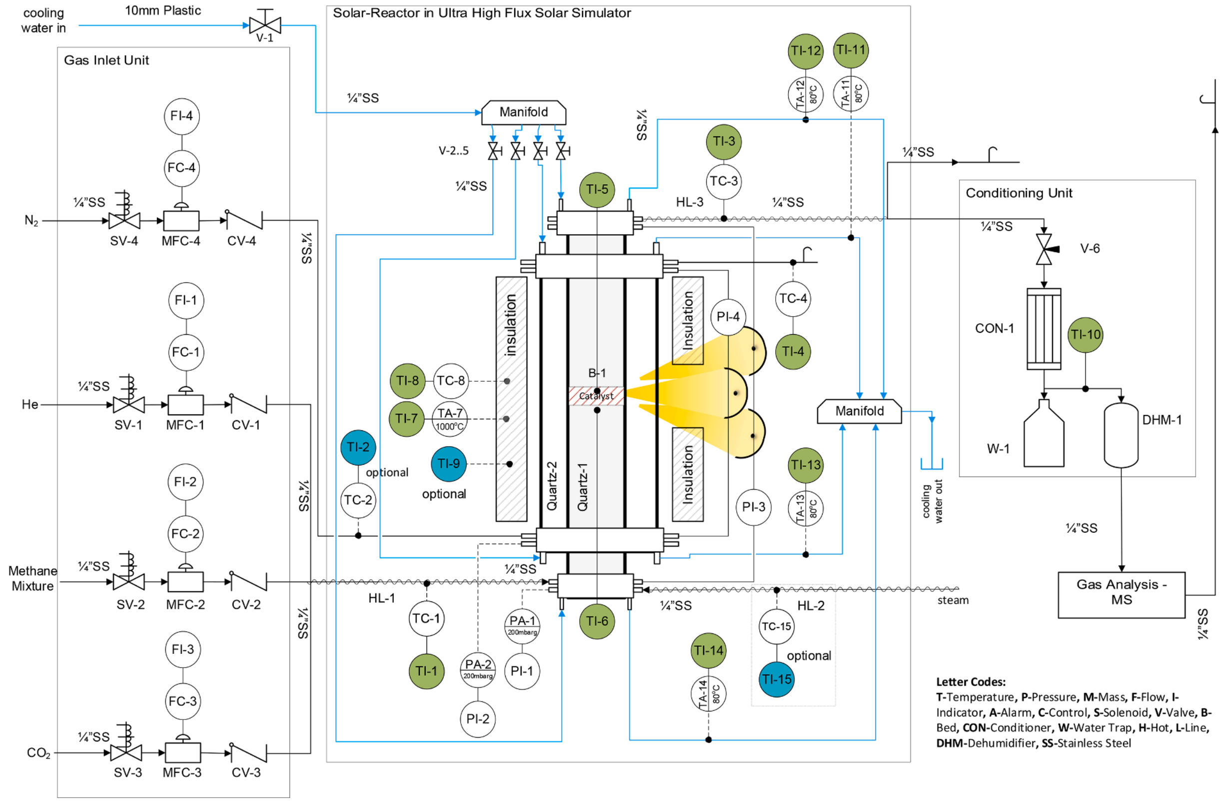

Fig. 3. Process and instrumentation diagram (PID) of the experimental reactor system.

Several proactive measures were taken for the safe operation of the system. In the case of pressure (relative or differential) exceeding 200 mbar, the mass flow controllers were reset, and the appropriate gas valves shut off. Threshold limits were set for the temperature sensors i.e.,1,000 °C for the bed, 800 °C for the cavity's wall, 400 °C for the gases' outlet, 120 °C for the gases before the evolved gas analysis, 80 °C for the cooling water's outlet, 70 °C for the reflectors' surfaces, and 80 °C for the inside of the HFSS enclosure. The interlock between the HFSS and reactor systems automatically enforced the same safety limits for either system. Two remotely controlled cameras viewed the reactor system's front and back sides. In addition to the building's flammable gas, oxygen, fire, and water alarm sensors three portable water leak, methane, and smoke detectors were placed below and above the reactor system respectively.

The composition of the effluent gases was monitored continuously using a mass spectrometer (HPR-20 by Hidden Analytical, UK) equipped with a heated (180 °C) capillary inlet. For the conversion of the partial pressures to molar ratios, the relative sensitivities of the concerned gases (CH_4, CO, CO_2, H_2, and H_2O) were estimated relative to He using for calibration a calibration mixture of 10 % CH_4, 10 % CO, 10 % H_2, 10 % CO_2 and 2 % Ar with He balance (accuracy 5 % of reported values by National Industrial Gas Plants, Qatar). The Ar ratio in the reacting and calibration standards served as an additional metric for the overall experimental uncertainty.

2.3. Irradiance measurement

For the irradiance on the solar reactor, a combination of flux-mapping and heat flux measurements was applied; more details are given elsewhere [52]. In brief, the mapping of the irradiance on the target was a three-step method known as flux characterization [44,53], which allowed for the determination of the spatial flux distribution with high accuracy and minimum direct measurements. At the target area, a Lambertian reflector (250x250 mm^2 stainless steel slab plasma coated with Al_2O_3 made to order by Haueter Engineering, Switzerland) was mounted on a 3-axis slider (NLS8–500–101, NLS8–300–102, and NLS8–200–101 precision sliders by NewMark) with a resolution of 0.08 µm and a 1 µm repeatability. The Lambertian reflector ensured that the incident light was reflected equally in all directions and the illuminated target appeared equally bright irrespective of the viewing angle [54]. A monochrome camera with an 8-bit and 1/1.8" charged-coupled device (CCD) sensor (1600x1200 DMK 51BG.02.H by TheImagingSource) with a 75 mm lens (C7528-M by Pentax) was used to capture images of the Lambertian target. Photos of the target under different illumination conditions were collected and processed to synthesize a complete flux map with arbitrary units (e.g., grayscale).

Similarly, a Gardon-type circular foil radiometer (heat flux gage by Vatell: TG1000-0 with a colloidal graphite coating, calibrated with 1 % repeatability, 2 mV (W cm^{-2})$^{-1}$ sensitivity, and a transducer calibration accuracy of ± 3 %) was installed at the center of the Lambertian target to measure the irradiance at different locations. The heat flux gage output on the order of 0–10 mV was logged using a 24-bit thermocouple input module (NI-9211 by National Instruments) at high resolution and voltage (range ± 80 mV) reading mode. The Lambertian target and flux gage were water-cooled at 25 °C (using an F50-ME chiller by Julabo) and equipped with multiple K-type thermocouples (OD 3 mm, IEC60584 Class 1: ~± 1.0 °C by Omega Engineering Inc UK) to monitor their temperature and prevent overheating.

Then the point heat flux measurements were combined with the flux map to obtain a high-resolution irradiance map. The power input to the reactor was obtained by integration of the irradiance maps over the aperture of the secondary concentrator.

2.4. Experimental design and performance metrics

Assessment of the solar thermochemical reactor performance was performed across four selected factors, i.e., temperature, gas flow rate, irradiation pattern, and catalyst reduction. The experimental design

(DoE) was structured as a fractional factorial: full for the first two factors and fractional for the rest. Table 1 presents the experimental factors and their selected levels (L#); note the examples provided as table notes. Table 2 presents the index of the selected experiments included in this analysis; the index number assists in identifying the results and discussion per experiment (E#). Important to note that the prescribed temperatures in Table 1 were achieved by utilizing the central lamp block and two or none of the side lamp blocks. In particular, the three irradiation modes (Smooth, Standard, and Focused) represent qualitatively the ratio of the irradiance provided by the side lamp blocks over the central one, with an empirically selected threshold of 0.14. When the ratio was larger than 0.14, the reactor's temperature appeared to be equally sensitive to equal incremental changes of the side and central lamps' positions. In contrast, for a ratio smaller than 0.14, the reactor temperature was mainly sensitive to changes of the central lamp position. As an illustration, for a ratio equal to zero, only the central block was used (i.e., Focused irradiation mode). For a ratio greater than 0.14, all three blocks were used (i.e., Smooth irradiation mode) but the side blocks provided more energy compared to a smaller ratio (i.e., Standard irradiation mode). To point out, the terms "Smooth" and "Focused" were selected posterior following the ray-tracing modeling of each configuration – partially discussed in Chapter 4.

In the literature, the performance indicators for solar-driven thermochemical processes vary with little consensus. Recent efforts aimed to unify the approaches by suggesting indicators for the most common applications [55]. For this reason, the performance indicators and their calculation follow the recommended approach by Gokon et al. [7].

Methane conversion, X [-], was estimated on the basis of synthesis gas production (i.e., the objective of the studied process) instead of methane consumption:

$$X = \frac{y_{CO} + y_{H_2}}{4y_{CH_4} + y_{CO} + y_{H_2}} \tag{1}$$

where y is the mole fraction of the respective component in the effluent gas.

First, the conversion of (solar energy) light to chemical energy, in other words, the chemical storage efficiency, η_{chem} [-], was estimated by assuming the combined dry methane reforming reaction:

$$CH_4 + (1 + x)CO_2 \rightleftharpoons (2 + x)CO + (2 - x)H_2 + xH_2O_{(g)} \quad \Delta H_{298K(overall)} = 247 + 41 \times x \text{ [kJ mol}^{-1}] \tag{2}$$

where × indicates the extent of the (reverse water–gas-shift) reaction, determined from the experimentally measured composition of the effluent gas. Second, the power stored as chemical enthalpy, W_{chem} [kW], was determined by:

$$W_{chem} = f_{CH_4} \times X \times \Delta H_{298K(overall)} \tag{3}$$

Table 1
The selected experimental factors and their operational levels.

Factor	Level 1 [L1]	Level 2 [L2]	Level 3 [L3]
Temperature [°C]	550	650	800
Gases Flow Rate (5 % CH_4, 5 % CO_2, 2 % Ar, He) [l min^{-1}]	0.58	0.85	1.94
or Gas Hourly Space Velocity (GHSV)			
[l $hour^{-1}$ $g_{cat.}^{-1}$]	17	26	58
[$hour^{-1}$]	865	1268	2895
Catalyst Reduction* or Oxidation**	Reduction	–	Oxidation
Irradiation Modes	Smooth (3 lamps)	Standard (3 lamps)	Focused (1 lamp)

* Catalyst was reduced with the produced hydrogen by maintaining the catalytic bed at 800 °C (maximum temperature) and 1.94 l min^{-1} (maximum flow rate) for one hour. A feed of hydrogen gas was not introduced for safety reasons.
** By purging with a 100% CO_2 flow instead of He.

where f_{CH4} [mol/s] is the feed molar flow rate of methane. Finally, η_{chem} [-] was calculated as the ratio of the chemical enthalpy (stored power) to the total power of the incident light, $W_{\text{inc,ap}}$ [kW], on the aperture of the reactor:

$$\eta_{chem} = \frac{W_{chem}}{W_{inc,ap}} \tag{4}$$

Using the above quantities, a series of other efficiency metrics can be derived, with a few presented in the following two chapters.

The total power of the incident light on the reactor aperture, $W_{\text{inc,ap}}$ [kW], was estimated by integrating the measured irradiance (see section 2.3). In essence, the above light-to-chemical energy conversion efficiency, η_{chem} [-], rightfully neglects the optical efficiency of the HFSS because such characterization is beyond this work's aim. On the other hand, η_{chem} inadvertently combines the contributions of the (secondary and tertiary) concentrators and the directly irradiated bed. To explain the importance of each reactor's component, and especially to investigate the benefit of the direct irradiation hypothesis, the respective contributions were inferred by optical modeling (section 2.1 describes the computer tool). Namely, the processing of the simulations of the optical model produced the percentage and distribution of the power incident on the catalyst bed. This power was expressed as $W_{\text{inc,bed}}$ [kW] and depended on the specific lamp(s) position and the cavity's walls temperature (reradiation). The lamps' positions relative to the reflector varied among the experiments because they were the main independent variables in controlling the reactor's temperature. The cavity's wall temperature was measured at each experiment and was introduced accordingly in the optical model. Eventually, an adjusted light to chemical energy efficiency, η'_{chem} [-], was estimated by introducing $W_{\text{inc,bed}}$ in Eq. (4) in the place of $W_{\text{inc,ap.}}$:

$$\eta'_{chem} = \frac{W_{chem}}{W_{inc,bed}} \tag{5}$$

The theoretical conversion of the DRR at equilibrium was estimated with the FactSage© (version 8.0) computer software that performs thermodynamic analyses; other alternatives are HSC Chemistry© and Aspen Plus©. FactSage© is thermodynamic modeling software that incorporates one of the largest fully integrated database computing systems in chemical thermodynamics. It accesses both solution and pure compound databases and it contains special calculation modules for different applications (equilibrium, phase diagram etc). The EQUILIB module applies the Gibbs energy minimization method. It calculates the concentrations of chemical species when specified elements or compounds react or partially react to reach a state of chemical equilibrium [56,57]. Over the years, many studies have used this software to simulate various processes such as H_2 production from biomass anaerobic gasification [58], H_2 production from homogeneous HI decomposition [59], H_2 production from hybrid water-splitting cycles [60,61], syngas production via chemical looping [62,63], and methane dry reforming [64]. In this study, the database FactPS (pure substances version 7.2) was used for the properties of the related compounds while the EQUILIB module was utilized for the thermodynamic calculations. The EQUILIB module offers the option to conduct a full exploratory investigation of the reaction products or select specific ones based on the reaction and objectives. In this study, the equilibrium calculations were repeated for all possible combinations of products in order to include the effect of carbon and/ or water formation and capture the minimum and maximum syngas production and H_2/CO ratios at equilibrium.

2.5. Materials and characterization

The catalyst bed consisted of 0.2 g of 20 wt% Ni on γ-Al_2O_3 support (Item #: 0040-ALNiA20 by Riogen Inc. US) premixed with 2 g of inert quartz sand. It had around 2 mm thickness, it was positioned at the center of the reactor's cavity (reacting zone), and it was kept in place using two quartz wool layers (99.995 % SiO_2, fibers of diameter in the

Table 2
Experimental conditions/factors for selected experiments (E#). The operational levels (L#) are as per Table 1*.

Factor	Index of Experiment (E#)														
	11	12	13	14	15	17	18	19	20	21	22	23	24	25	26
Temperature [°C]	L3	L3	L2	L2	L2	L1	L3	L1	L1	L2	L3	L2	L2	L2	L2
Gas Hourly Space Velocity (GHSV)	L2	L3	L1	L2	L3	L1	L1	L2	L3	L2	L3	L2	L2	L1	L2
Catalyst Reduction, Oxidation, or Not											L1		L1	L1	L2
Irradiation Mode	L1	L2	L2	L2	L1	L2	L1	L2	L2	L3	L1	L2	L1	L2	L2

* Example A: The index E22 points to experiment 22 with experimental levels L3 = 800 °C for temperature, L3 = 2895 h^{-1} for GHSV, with Reduced catalyst (L1) and Smooth irradiation mode (L1), as per Table 1. **Example B:** The index E14 points to experiment 14 with experimental levels L2 = 650 °C for temperature, L2 = 1268 h^{-1} for GHSV, with no additionally treated catalyst (–) and Standard irradiation mode (L2), as per Table 1.

range of 5–40 μm by Robson Scientific UK) and the bottom reactor thermocouple (see Fig. 2**C**). After each experiment, the catalytic bed and quartz wool were extracted from the inner tube and were sieved to a 53–250 μm size range before characterization. The crystalline phases of fresh and spent catalysts were analyzed by X-ray diffraction (powder XRD) on a diffractometer (Ultima IV Multipurpose X-ray by Rigaku) using a Cu-Kα radiation (λ = 1.5406 Å) source equipped with a fixed monochromator, cross beam optics, and a scintillation counter. The XRD diffractometer was operated at 40 kV and 30 mA using divergence (2/3°), divH.l. (10 mm), scattering (1.17 mm), and receiving (0.3 mm) slits. The XRD data were collected using continuous scan mode in the 2θ range of 10–80° with a step width of 0.02° and 1°/minute scan speed. The morphological characterization of the catalyst particles was conducted by high-resolution environmental scanning electron microscopy (ESEM) on a microscope (Quanta 400 by FEI) operated at high vacuum mode with 5 kV acceleration voltage and a working distance of around 5 mm. Chemical analysis of the catalysts was carried out by the energy-dispersive X-ray spectroscopy (EDS: INCAx-sight liquid nitrogen cooled detector with a Si (Li) window). Oxygen temperature programmed oxidation (O_2-TPO) was conducted to estimate the carbon deposition on the catalyst by flowing 10 % O_2 (He balance) at 50 ml min^{-1} from room temperature to 800 °C with a heating rate of 10 °C min^{-1} on a thermal analyzer (Q600 SDT by TAInstruments).

2.6. Experimental uncertainty

All instruments or procedures that could be calibrated before or during the experimental campaign had been calibrated or were purchased with a calibration certificate. Duplicate sensors were employed for measurements, like the reactor temperature measurement with two thermocouples (TI-5 & 6 in Fig. 3), where calibration within this experimental campaign was not possible, although the respective vendors have provided uncertainty ranges. All measuring uncertainties have been provided within the text in the earlier paragraphs. In the context of this work, the two main experimental quantities were the evolved gases concentrations and the irradiance levels to the reactor surface. The former were used to calculate the reactants' conversion efficiencies. Therefore, the experimental uncertainty was estimated as the standard deviation among repeatable tests and equaled 3 % of the reported values. At the same time, the irradiance on the aperture of the secondary concentrator was measured with a 3 % uncertainty. Following standard principles of error analysis, the overall uncertainty of the energy efficiency calculations was estimated to be around 10 % of the reported values in the results. Note that the irradiance on the reactor's periphery was not measured but was defined computationally with optical modeling (see Section 2.1 and Chapter 4). For this reason, the uncertainty for the adjusted energy efficiency could be higher than for the standard energy efficiency.

3. Results and discussion

The design of the reactor parts was supported by optical modeling, especially the secondary and tertiary concentrators, assuming that up to three (the central and two side) lamp blocks would have been adequate to achieve the experimental conditions. When all three lamp blocks were fully focused on the catalytic bed (not on the aperture), a uniform flux distribution was achieved on the 50 mm radius aperture (Fig. 4**A**). Indicatively, the irradiance drops just by 50 % within 80 % of the aperture's area (up to 40 mm radius) which is an outcome of the Gaussian profile of the concentrated light. Under these conditions, the irradiance on the catalyst bed reached higher values (up to three times more; Fig. 4**B**) as expected since all three lamp blocks were focused on the bed. Despite the design efforts to maintain a uniform irradiance distribution on the bed, the ratio between the front and back irradiance levels was around five (Fig. 4**B**). The confidence in the optical modeling results is high because of the excellent comparison between the measured and modeled irradiance levels at different lamp positions (Fig. 4**C**), as described in Section 2.6 and earlier works [52].

All methane reforming results presented herein were produced with the same experimental procedure. At first, the reactor was purged with a 20 % higher flow rate of He than the one prescribed in DoE. At the same time, it was illuminated to achieve and maintain the reactor temperature for at least 30 min. By manually regulating the relative position of the lamps to the reflectors, it was possible to reach the DoE temperature within 2–5 min compared to the few hours required for a typical furnace with the same nominal electrical power; based on own calculations. Then, the reacting mixture was flown into the reactor by replacing the He flow. By the previous empirical procedure, the reactor temperature drop, owing to the endothermic DRR reaction, was kept to less than 50 °C, which alternatively could drop more than 100 °C. The overall behavior is illustrated in Fig. 5 for the selected time series of the syngas conversion and the bed temperature (reacting mixture injection at 0 min). The evolved gases and reactor conditions were monitored continuously, as described earlier, and maintained at least for 60 min after reaching approximately 98 % of the maximum H_2 fraction. Beyond the initial temperature drop, the temperature was maintained at the predefined level with a standard deviation better than 3 °C.

Fig. 6 summarizes the results of the experimental campaign (Table 1 and Table 2). The overall patterns concerning CH_4 conversion and the molar ratio of the produced H_2 to CO were the expected ones. At the highest temperature (L3: 800 °C), conversion to syngas (Eq. (1); Fig. 6**A**) approached 93 % close to the maximum equilibrium levels according to the thermodynamic calculations, irrespective of the GHSV values and within the experimental uncertainty (±3%). At the medium temperature (L2: 650 °C), the maximum conversion was achieved for the middle GHSV. The experiments with the mildly reduced catalyst (bars with linear pattern) also reached the same maximum conversion. On the other hand, at the higher GHSV the conversion reduces slightly or more compared with the lower GHSV. It is an expected pattern attributed to the critical role of the reactants' residence time and the kinetics of the catalytic reaction [32]. Conversely, the lowest conversion at around 26 % was observed for the highest GHSV and lowest temperature (L1: 550 °C). The conversion increased progressively for the lower GHSV values approaching 45 % for the lowest one. No experiments were conducted with a mildly reduced catalyst at the lower temperature level but a similar pattern to the middle-temperature level is expected.

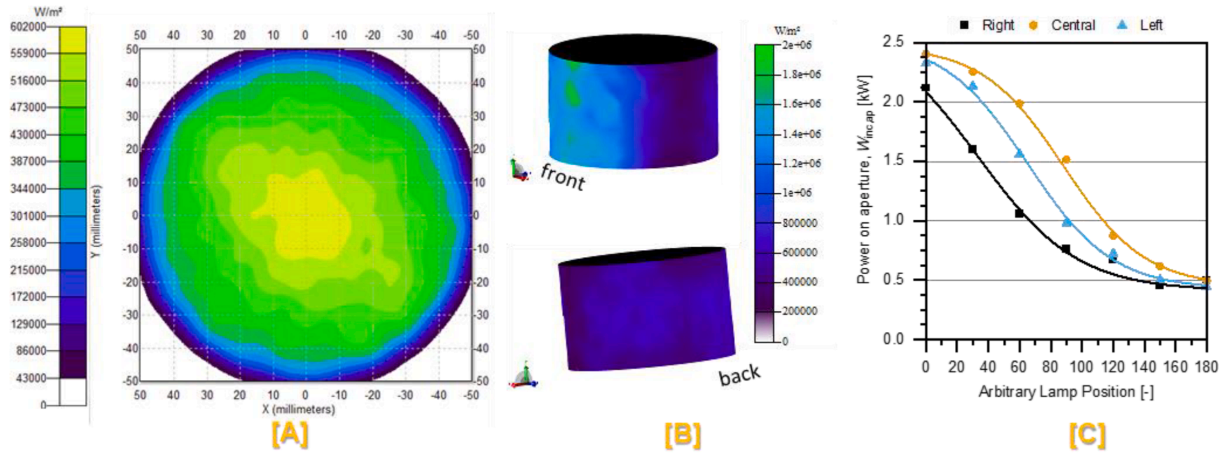

Fig. 4. Modeled flux maps on A) the aperture of the reactor and B) the catalyst bed with three lamp blocks (Right, Central & Left) active and targeting the catalyst bed, and C) comparison of the measured radiative power on the aperture (closed symbols) with the modeled power (continuous lines) at different lamp positions (with 0 being fully focused) for each of the three utilized lamp blocks while targeting the aperture.

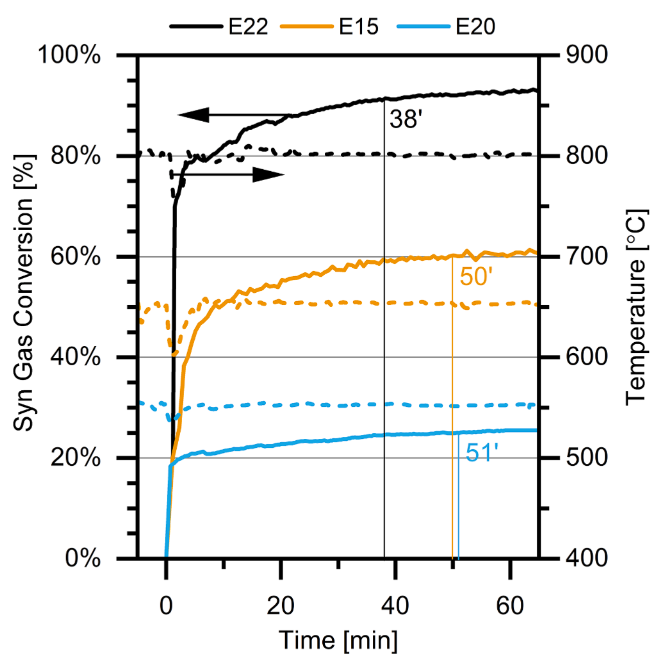

Fig. 5. Indicative conversion (left axis and solid lines) and temperature (right axis and dashed lines) profiles for the three temperature levels (E#) towards the steady state. The reacting mixture was injected into the reactor after achieving the steady state temperature – set as 0 min. The vertical lines denote the time to reach 98 % of maximum conversion.

Furthermore, it is possible to deduce or presume which DRM-related reactions were favored, at each temperature and GSHV, by comparing the experimental conversion rates with the equilibrium ones for the different reaction combinations. However, this is beyond the scope of this study; nevertheless, an interested reader can check related works e.g. on the thermodynamics [65] or thermokinetics [66] of the DRM.

Compared to the significant variations of the syngas production (i.e., conversion of methane and carbon dioxide to hydrogen and carbon monoxide) with the different GSHV, temperatures, and irradiation modes, the corresponding H_2/CO ratios (±0.025 estimated uncertainty) varied in a relatively more minor range of less than 9 %, at each temperature level (Fig. 6**B**). The ratios were close but below the lower equilibrium values for the experimental runs at 650 °C and 800 °C and slightly above at 550 °C. The former observation can be attributed to the kinetically suppressed carbon formation and the enhanced carbon dioxide conversion (owe to increased consumption in the RWGS under the experimental conditions. The observation is also consistent with earlier theoretical [65] and experimental works [67], which clearly state the carbon formation dependence on the temperature and catalyst.

The solar-to-chemical energy efficiencies, η_{chem}, by utilizing the measured incident power on the reactor aperture were estimated to be less than 1 % in all experimental runs. These indicate the large amounts of energy lost at the different reactor components (i.e., secondary and tertiary concentrators, reradiation, cooling flanges, and insulation). Therefore, the reactor and concentrators need to be redesigned and further insulated to approach efficiencies reported elsewhere, e.g., greater than20 % by Gokon et al. [7] Gokon et al. [7]. As a matter of fact, the optical modeling of the as-built solar reactor configuration inferred that only 1 %-6% of the incident power on the reactor aperture reached the bed. This was attributed to two inadvertent deviations from the experimental design: i) the manufactured tertiary concentrator deviated from the computer design because of a CNC equipment constraint, and ii) the tertiary concentrator was designed assuming that the utilized lamp blocks would operate at a fully focused position – which eventually was not necessary. Following this observation, the adjusted energy efficiency, η'_{chem}, was considered more appropriate to examine this work's hypothesis. Most experiments yielded efficiencies in the range of 6–13 % (Fig. 6**C**) with some of them reaching more than 25 %. The next chapter performs a detailed analysis of the optical modeling results to investigate the impact of the reactor components' optical characteristics and the irradiation modes on the overall conversion efficiency.

The XRD patterns of fresh and spent catalysts are presented in Fig. 7**A**. The pattern of the fresh catalyst includes prominent peaks of the support γ-Al_2O_3 (peaks α and θ) and metal Ni (peak ζ), as expected. Additionally, there is an indication of oxidized Ni (peaks η and ι). The analysis of the spent catalyst XRD patterns proved challenging because of the large amounts of quartz (sand-filler and wool-bed support) despite the efforts to sieve them out. In particular, although the features in the XRD pattern of quartz do not have significant overlaps with those in the other patterns, the large amounts of quartz reduced the concentrations of other crystals and affected their signal-to-noise ratio. Hence, it sustained their identification difficult. In some cases (e.g., E26), the removal of quartz was more successful without an apparent reason. Nonetheless, it is still possible to identify the Ni (peak ζ) but not anymore the NiO (peak η) in most of the spent catalysts (e.g., E12, E18 and E23 in Fig. 7**A**), an observation attributed to the reducing conditions due to H_2 production. On the other hand, the XRD pattern of E26, which concluded with a mild oxidation step with CO_2, does not include the Ni features (peaks γ and η) but solely those of γ-Al_2O_3 (peaks α, δ, and θ) and NiO (peaks β, η, and ι). In any other case, including those not

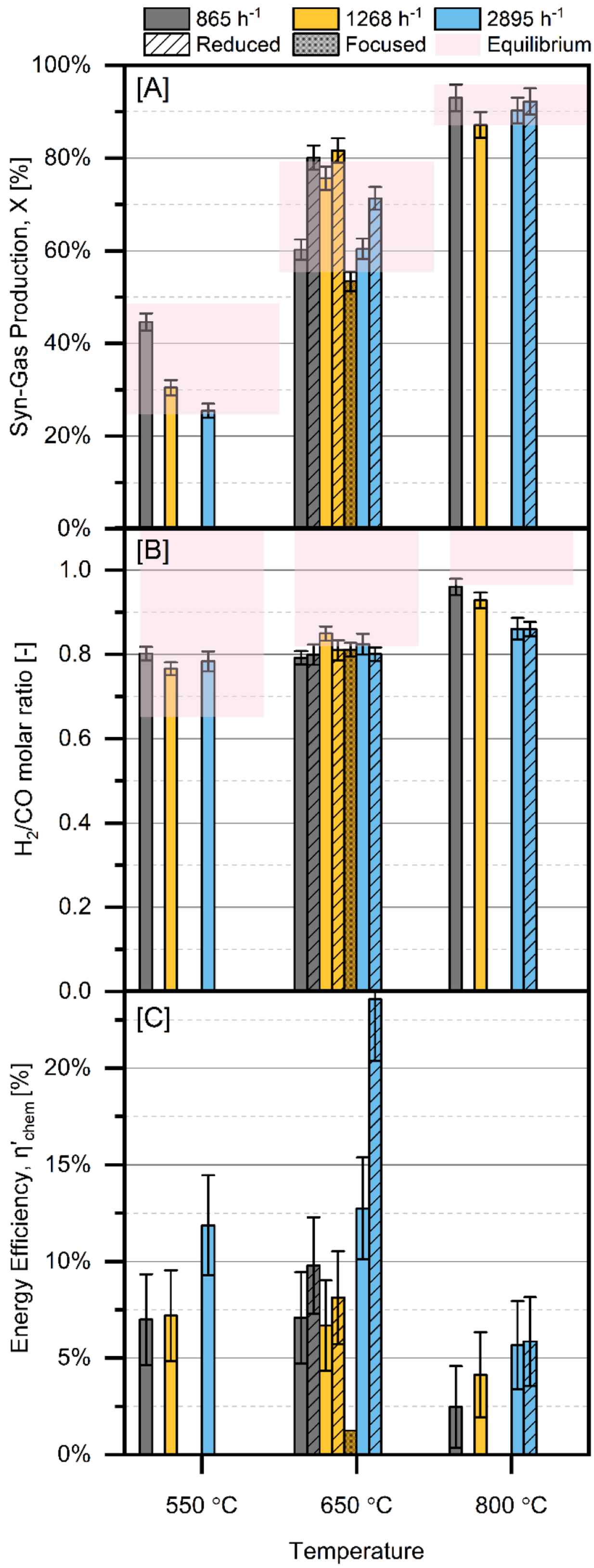

Fig. 6. Experimental results for the three temperature levels and space velocities (colored bars), including the thermodynamic equilibrium calculations (shaded areas), for A) the syngas production (as per Eq. (1)), B) the ratio of produced H_2 to CO, and C) Incident-to-H_2 adjusted energy conversion efficiency (as per Eq. (5)). The charts include the experiments with reduced catalyst (bars with the linear pattern) and focused irradiance (bars with the dotted pattern).

presented herein, there seems to be no significant difference among the collected crystallographic patterns.

The presence of quartz rods in the samples was verified in the SEM images of all spent catalyst samples (e.g., Fig. 7**B and D**). The SEM images showed aggregation to larger particles with increasing temperature e.g. by comparing E14 at 650 °C (L2, Fig. 7**D**) with E11 at 800 °C (L3, Fig. 7**B**). The aggregation was attributed to high-temperature sintering because of the characteristic lack of rough features on the surface of catalyst particles (e.g., Fig. 7**E**). In many experiments, the surface of catalyst particles was covered entirely with carbon nanowires (filamentous carbon) with diameters between 50 and 100 nm and longer than a few micrometers (e.g., Fig. 7**C**). Synthesis of the SEM images from all experiments revealed that production of nanowires took place during all high-temperature tests (L3: 800 °C) regardless of the GHSV and during the combined low temperature (L1: 550 °C) and low GHSV (L1: 865 h^{-1}) conditions. In the remaining cases (e.g., L2 temperature and GHSV there was no production of filamentous carbon based on the optical analysis of the SEM images.

The verification of the type of carbon formed and quantification of the carbon formation rate were extracted from the O_2-TPO analysis. Selected O_2-TPO results of weight loss [%] and weight derivative w.r.t. temperature [mg °C^{-1}] are illustrated in Fig. 8. In particular, three types of carbon formation peaks were noted in the weight derivative graph based on the temperature of appearance: 450–520 °C, 520–570 °C, and 670–630 °C. According to the extensive literature on the type of carbon formed on Ni-based catalysts (e.g., Kumar et al. [69], Trimm [70]), the first and second types were assigned to amorphous carbon or a mixture of amorphous with atomic carbon, and the third type to filamentous carbon. The assignments agree with the SEM images and other experimental campaigns with the same batch of catalyst [71]. The total carbon mass formed increased with the decrease of temperature and GHSV (e.g., carbon amounts E17 > E19 > E14 > E11 in Fig. 8). The estimated carbon formation rates appear to be almost half of the ones reported in Chatla et al. [71] for the same batch of catalyst and temperature when tested in a typical bench scale microreactor (e.g., at 650 °C ~ 11.9 × 10^{-4} mg C h^{-1} g^{-1} vs 6.4 × 10^{-4} mg C h^{-1} g^{-1} here), however, with a reacting gas mixture twice as rich in CH_4 and CO_2.

4. Impact of the irradiation patterns

The earlier discussion on the experimental results showed a dependence on the different irradiation patterns. Therefore, optical modeling was further employed to analyze the various experimental configurations and especially the optical losses. Particularly the latter can offer valuable insights for future designs because of the directly irradiated bed, which is central and unique in this study. In parallel, the very low solar-to-chemical energy efficiency and its prominent dependence on the reference irradiance mark the importance of the irradiation pattern (η_{chem} less than 1 % for the reactor aperture irradiance vs adjusted η'_{chem} 5 %-25 % for the bed periphery irradiance). Furthermore, by intuition, the syngas production should be independent of the irradiation pattern under the same experimental conditions (temperature, GHSV, and irradiation level) because of the small size of the bed. Then again, as seen in Fig. 9**A** the syngas production efficiency dropped from around 77 % to less than 44 % as the irradiation pattern changed from smooth to focused for the same experimental conditions (L2 temperature and GHSV) and power levels (1.81 ± 0.08 kW). At this point, it is crucial to clarify the differences among the irradiation patterns.

According to the standard and empirical experimental procedure (e.g., see results for E14 in Fig. 9), the three lamps were ignited almost simultaneously while being mildly unfocused. Then the focus of the central lamp was increased gradually to reach the prescribed temperature. Whenever necessary, only the central lamp was utilized to adjust the irradiation and reactor temperature – referred to as Standard irradiation mode. In the Smooth irradiation mode (i.e., E24), all three lamps were adjusted in parallel to control the temperature. In the Focused

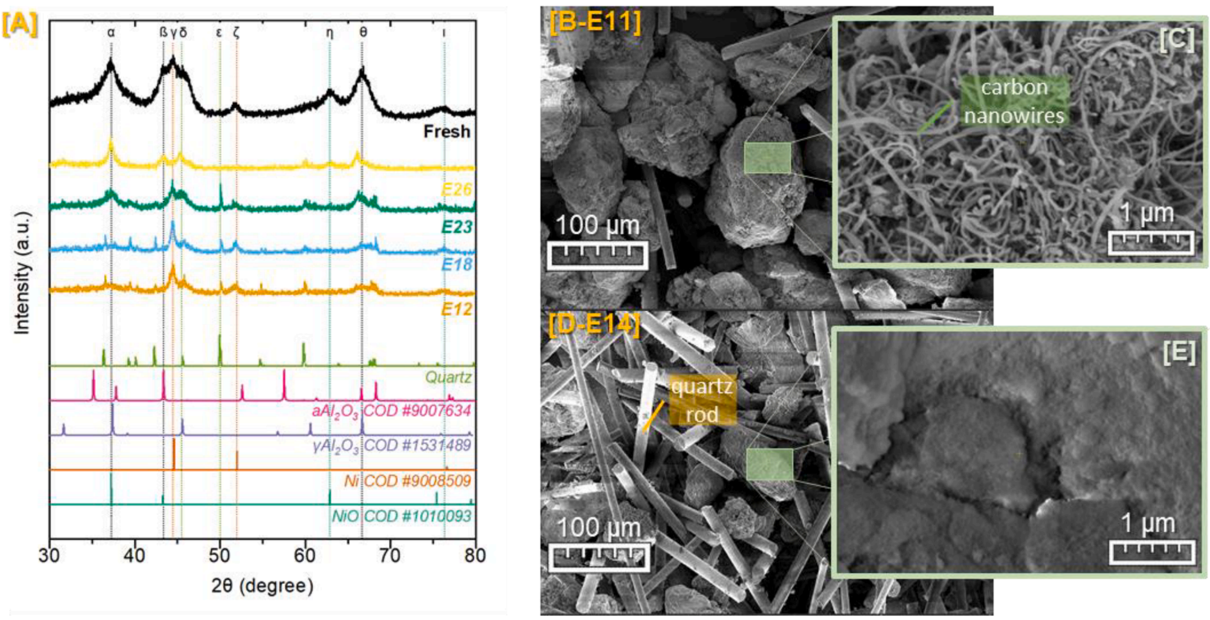

Fig. 7. Results of the catalyst characterization: A) XRD patterns of the fresh and spent catalyst at different experiments (E#), including reference powder diffraction cards (Crystallography Open Database [68] and B-E) SEM images for selected experiments (E#).

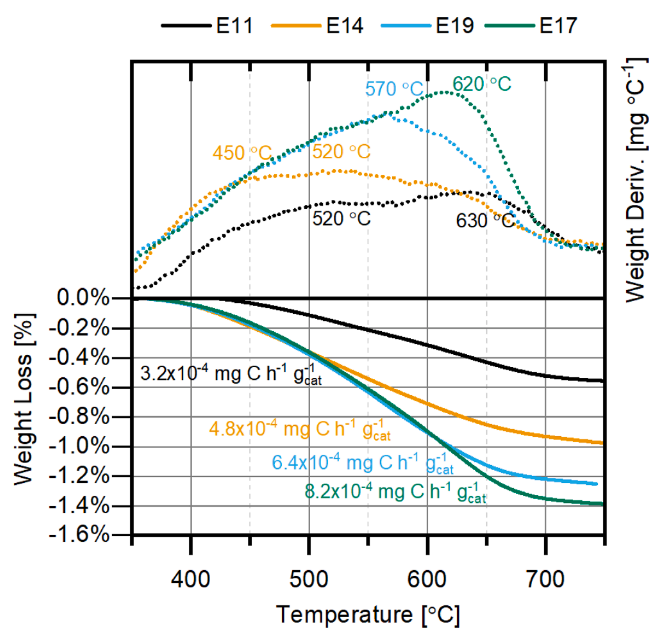

Fig. 8. Results for the O_2-TPO weight loss [%] and weight derivative w.r.t. temperature [mg °C^{-1}] of selected spent catalysts (E#). The figure includes the peak temperatures (based on the weight derivatives) and the average mass production of Carbon per hour, and the mass of the catalyst [mg C h^{-1} g_{cat}^{-1}].

irradiation mode (i.e., E21), only the central lamp was utilized, whereas it was not necessary to operate it perfectly focused. At steady state, the ratio of the power delivered from the central lamp to the power delivered from the side lamps varied from ~ 0.14 to 0.07 and 0.0 for the Smooth, Standard, and Focused irradiation patterns respectively (Fig. 9). Earlier in the description of the reactor design via optical modeling, it was mentioned that the optimization of the tertiary concentrator was carried out assuming all three lamps were placed at the optimal focus position (i.e., fully focused). An assumption based on past literature and experience ignoring that lower irradiance levels may have been adequate to achieve the reaction temperatures. The impact of this design aspect is illustrated in Fig. 9**B** by the almost 8x times higher irradiance levels on the front side of the bed compared with the lateral and back sides. Notwithstanding the smaller amount of power intercepted by the bed while in Smooth mode, the adjusted energy conversion efficiency was higher than the other modes, also driven by the more significant syngas production.

To answer the above questions with quantitative means, 3D heat transfer microscopic modeling must be performed, which was beyond the scope of this work. Qualitatively, our main hypothesis is that the different irradiance patterns affect the catalytic bed's internal temperature distribution, with the Smoothed pattern approaching uniformity. Although the temperature at the bed's core was monitored and the bed has a small volume, the non-uniform irradiance creates hot and cold spots on the surface of the catalytic bed which subsequently create hot and cold spots within the bed. Literature and own observations support the considerable sensitivity of the studied reactions to minor temperature variations. Hence, a non-uniform temperature bed with cold spots will yield lower amounts of synthesis gas and reduce the system's energy efficiency. At the same time, Fig. 9**B** shows a reduction of the energy delivered on the catalytic bed for the Focused irradiance mode which increases the energy demand to achieve the required temperature and subsequently further reduces the system's energy efficiency.

Unfortunately, insufficient data exists to attempt a deeper phenomenological interpretation of these observations. To emphasize, the experimental conditions across the three operational modes were as close as possible and within the experimental uncertainty. Nevertheless, a conceptual energy flow was devised (Fig. 10) based on the optical modeling of the various experimental lamp-block configurations and the average conversions of methane and carbon dioxide to synthesis gas. Overall, the optical losses at the primary reflector (70 %) and the secondary & tertiary concentrators (98 %) appear to have the most significant impact, with the latter attributed to the design issue discussed earlier. Without a doubt, the latter optical losses can be significantly reduced with a better design of the reactor's cavity. Similarly, other heat losses (~59 %) can be reduced by improving the insulation and recovering the heat of the exhaust gases (~32 %).

5. Conclusions

Following the detailed presentation and discussion of the results

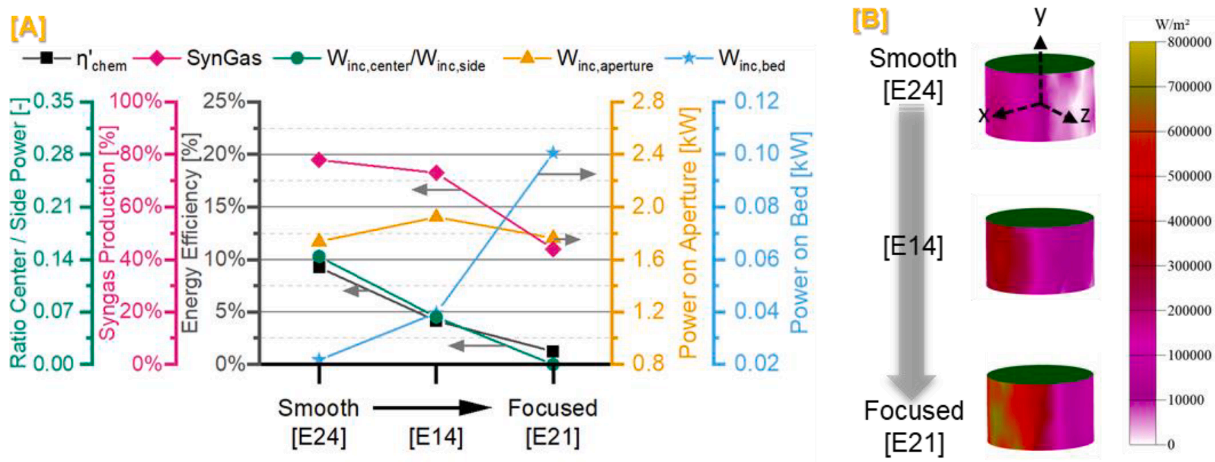

Fig. 9. Analysis of the energy delivery strategy for three selected delivery strategies (from Smooth to Focused; all at 650 °C and 1270 h^{-1}) with A) showcasing the variation of the energy-efficiency (experimental; black squares), syngas production (experimental; magenta diamonds), transmitted power via the reactor aperture (front; 50 mm diameter; orange triangles), delivered power on the catalytic bed (blue stars), and the ratio of bed irradiance sources (central lamp/side lamps; green circles); and B) illustrating the 3D irradiance maps on the catalytic bed periphery (sources direction on -x). The vertical axes ranges were selected to encompass the extent of the observed values in the experimental campaign.

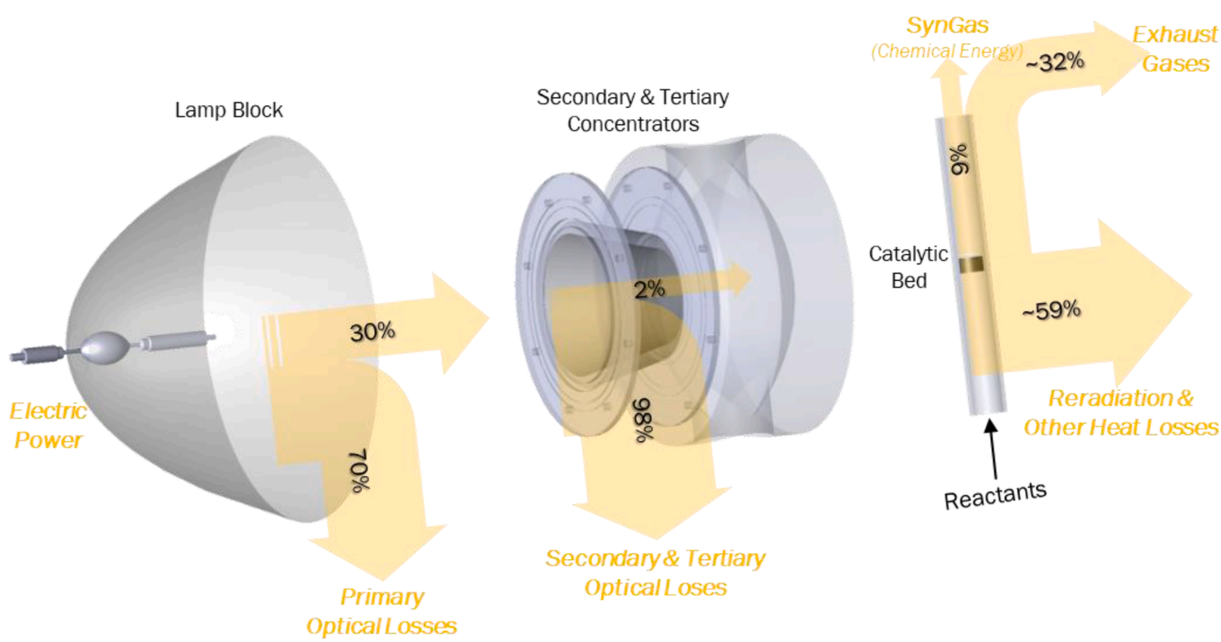

Fig. 10. Conceptual energy flow diagram based on the optical modeling calculations and the measured conversion of reactants to synthesis gas.

collected in this study, the conclusive points can be grouped into three main categories. The first category relates to the observations supporting the central hypothesis of this work. Indeed, the potential to drive the dry reforming reaction of methane and carbon dioxide in a tubular solar reactor with a directly irradiated catalyst bed was clearly demonstrated. The reported syngas production efficiencies and quality (ratio of H_2 to CO) were on par with the literature. Furthermore, the inner quartz tube showed no signs of degradation despite more than 80 h of intermittent testing. Although, this was one of the main threats identified in the Hazard Identification stage (HAZID) of the experimental design. The inner quartz tube had to be replaced a few times during the first days of the experimental campaign because of breakage at the mounting points. Only once the quartz tube was replaced because of carbon (graphite) layer formation at the front side of the catalyst bed. At that time, all three lamps were ignited while fully focused, resulting in a bed temperature exceeding 1,300 °C in just 20 s.

The second category relates to erroneous experimental assumptions and design decisions that may have affected the spent catalyst characterization and definitely affected the energetic quantitative estimations. With a maximum adjusted light-to-chemical energy efficiency of around 25 %, the performance of the solar reactor was similar to other literature systems. On the other hand, the very low actual efficiency was attributed to the optical performance of the concentrators, according to the optical model's analysis. Nonetheless, the combination of the experimental results with the optical modeling ones pointed out that smooth irradiance levels enhanced the energy and syngas production efficiencies compared to the focused levels and regardless of the small bed diameter. Concerning the carbon formation, analysis of the spent catalyst yielded comparable to the literature mass production rates and types, although the overall amounts were small because of the short duration experiments and challenging to separate the catalyst from the filler material. The latter can possibly be addressed with the use of a larger size filling material and bed support than the sand and quartz wool employed here respectively.

Expressively, many of the conclusive points in the second category spawn in the third category, which addresses potential areas of future investigation and interesting new directions. The control of a high flux solar simulator is challenging and heavily relies on the user's experience. Although the temperature, as the process variable of interest, can remain stable among reasonable limits, the highly sensitive homogeneity of the irradiance levels can affect the performance and create runaway conditions. A standard automated control system based on data-driven techniques or feedback loops may be adequate to control the temperature. However, it will require extensive investigation and additional process input to avert the latter. Finally, future experimental and computational work should investigate the impact on the performance and sensitivity of the bed diameter and the application of longer and parallel beds. Noteworthy, the proposed reactor configuration has the potential to host any length of reaction bed, with the appropriate adjustment of the light sources, although herein, only a fixed length of bed was demonstrated.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Konstantinos E Kakosimos reports financial support was provided by Qatar National Research Fund.

Data availability

Data will be made available on request.

Acknowledgments

This publication was made possible by two NPRP awards (experimental development, reactor fabrication, and data collection and analysis in NPRP X-100-2-024; thermodynamic calculations, ray tracing simulations, and interpretation of optical simulations in NPRP12S-0301-190191) from the Qatar National Research Fund (a member of The Qatar Foundation). Open Access funding provided by the Qatar National Library. The High-Performance Computing resources and services used in this work were provided by the IT Research Computing group at Texas A&M University at Qatar. The statements made herein are solely the responsibility of the authors.

References

[1] B. Wei, R. Fakhrai, B. Saadatfar, G. Mohan, T. Fransson, The Design of a Solar-driven Catalytic Reactor for CO2 Conversions, Energy Procedia 57 (2014) 2752–2761, https://doi.org/10.1016/j.egypro.2014.10.307.

[2] E.J. Sheu, E.M.A. Mokheimer, A.F. Ghoniem, A review of solar methane reforming systems, Int. J. Hydrogen Energy 40 (38) (2015) 12929–12955, https://doi.org/10.1016/j.ijhydene.2015.08.005.

[3] M. Steinberg, H.C. Cheng, Modern and prospective technologies for hydrogen production from fossil fuels, Int. J. Hydrogen Energy 14 (11) (1989) 797–820, https://doi.org/10.1016/0360-3199(89)90018-9.

[4] O. Faye, J. Szpunar, U. Eduok, A critical review on the current technologies for the generation, storage, and transportation of hydrogen, Int. J. Hydrogen Energy 47 (29) (2022) 13771–13802, https://doi.org/10.1016/j.ijhydene.2022.02.112.

[5] C. Graves, S.D. Ebbesen, M. Mogensen, K.S. Lackner, Sustainable hydrocarbon fuels by recycling CO2 and H2O with renewable or nuclear energy, Renewable Sustainable Energy Rev. 15 (1) (2011) 1–23, https://doi.org/10.1016/j.rser.2010.07.014.

[6] C.C. Agrafiotis, C. Pagkoura, S. Lorentzou, M. Kostoglou, A.G. Konstandopoulos, Hydrogen production in solar reactors, Catal. Today 127 (1–4) (2007) 265–277, https://doi.org/10.1016/j.cattod.2007.06.039.

[7] N. Gokon, Y. Yamawaki, D. Nakazawa, T. Kodama, Ni/MgO–Al2O3 and Ni–Mg–O catalyzed SiC foam absorbers for high temperature solar reforming of methane, Int. J. Hydrogen Energy 35 (14) (2010) 7441–7453, https://doi.org/10.1016/j.ijhydene.2010.04.040.

[8] D. Yadav, R. Banerjee, A review of solar thermochemical processes, Renewable Sustainable Energy Rev. 54 (2016) 497–532, https://doi.org/10.1016/j.rser.2015.10.026.

[9] S.A.M. Said, M. Waseeuddin, D.S.A. Simakov, A review on solar reforming systems, Renewable Sustainable Energy Rev. 59 (2016) 149–159, https://doi.org/10.1016/j.rser.2015.12.072.

[10] Z. Li, Q. Lin, M. Li, J. Cao, F. Liu, H. Pan, Z. Wang, S. Kawi, Recent advances in process and catalyst for CO2 reforming of methane, Renewable Sustainable Energy Rev. 134 (2020) 110312.

[11] H. Song, X. Meng, Z.J. Wang, H. Liu, J. Ye, Solar-Energy-Mediated Methane Conversion, Joule 3 (7) (2019) 1606–1636, https://doi.org/10.1016/j.joule.2019.06.023.

[12] M. Romero, A. Steinfeld, Concentrating solar thermal power and thermochemical fuels, Energy Environ. Sci. 5 (11) (2012) 9234–9245, https://doi.org/10.1039/C2EE21275G.

[13] C. Agrafiotis, H. von Storch, M. Roeb, C. Sattler, Solar thermal reforming of methane feedstocks for hydrogen and syngas production—A review, Renewable Sustainable Energy Rev. 29 (2014) 656–682, https://doi.org/10.1016/j.rser.2013.08.050.

[14] H.I. Villafán-Vidales, C.A. Arancibia-Bulnes, D. Riveros-Rosas, H. Romero-Paredes, C.A. Estrada, An overview of the solar thermochemical processes for hydrogen and syngas production: Reactors, and facilities, Renewable Sustainable Energy Rev. 75 (2017) 894–908, https://doi.org/10.1016/j.rser.2016.11.070.

[15] V.M. Wheeler, R. Bader, P.B. Kreider, M. Hangi, S. Haussener, W. Lipiński, Modelling of solar thermochemical reaction systems, Sol. Energy 156 (2017) 149–168, https://doi.org/10.1016/j.solener.2017.07.069.

[16] S. Chuayboon, S. Abanades, S. Rodat, Syngas production via solar-driven chemical looping methane reforming from redox cycling of ceria porous foam in a volumetric solar reactor, Chem. Eng. J. 356 (2019) 756–770, https://doi.org/10.1016/j.cej.2018.09.072.

[17] S. Chuayboon, S. Abanades, Thermochemical performance assessment of solar continuous methane-driven ZnO reduction for co-production of pure zinc and hydrogen-rich syngas, Chem. Eng. J. 429 (2022) 132356.

[18] J. Ma, B.o. Jiang, L. Li, K. Yu, Q. Zhang, Z. Lv, D. Tang, A high temperature tubular reactor with hybrid concentrated solar and electric heat supply for steam methane reforming, Chem. Eng. J. 428 (2022) 132073.

[19] Q. Zhao, B. Su, H. Wang, A. He, R. He, H. Kong, X. Hu, Mid/low-temperature solar hydrogen generation via dry reforming of methane enhanced in a membrane reactor, Energy Convers. Manage. 240 (2021), 114254, https://doi.org/10.1016/j.enconman.2021.114254.

[20] X. Shi, X. Zhang, F. Wang, L. Yang, Y. Dong, Y. Shuai, Thermochemical analysis of dry methane reforming hydrogen production in biomimetic venous hierarchical porous structure solar reactor for improving energy storage, Int. J. Hydrogen Energy 46 (11) (2021) 7733–7744, https://doi.org/10.1016/j.ijhydene.2020.12.034.

[21] Y. Dong, X. Zhang, F. Wang, G. Zhang, X. Shi, Y. Shuai, Numerical study on the thermal performance analysis of packed-bed latent heat thermal storage system with biomimetic vein hierarchical structure, Int. J. Green Energy 19 (6) (2022) 592–602, https://doi.org/10.1080/15435075.2021.1952208.

[22] X. Shi, F. Wang, Z. Cheng, H. Liang, Y. Dong, X. Chen, Numerical analysis of the biomimetic leaf-type hierarchical porous structure to improve the energy storage efficiency of solar driven steam methane reforming, Int. J. Hydrogen Energy 46 (34) (2021) 17653–17665, https://doi.org/10.1016/j.ijhydene.2021.02.171.

[23] X. Chen, F. Wang, Y. Han, R. Yu, Z. Cheng, Thermochemical storage analysis of the dry reforming of methane in foam solar reactor, Energy Convers. Manage. 158 (2018) 489–498, https://doi.org/10.1016/j.enconman.2017.12.066.

[24] H. Zhang, Y. Shuai, B.G. Lougou, B. Jiang, D. Yang, Q. Pan, F. Wang, X. Huang, Effects of foam structure on thermochemical characteristics of porous-filled solar reactor, Energy 239 (2022), 122219, https://doi.org/10.1016/j.energy.2021.122219.

[25] X. Shi, Y. Shuai, F. Wang, C. Zhang, Z. Cheng, X. Chen, Effects of ordered hierarchically porous structure on methane reforming performance in solar foam reactor, Journal of CO2 Utilization 37 (2020) 147–157, https://doi.org/10.1016/j.jcou.2019.12.002.

[26] H.S.-J. Long Hua-Li, Xu Yan, Qin Pan, Shang Shu-Yong, Yin Yong-Xiang, Dai Xiao-Yan, Catalytic Activity of Ni/MgO-Al₂O₃ Catalytically Activated Absorber for CO₂ Reforming with CH₄ Driven by Direct Light Irradiation, Chin. J. Catal. 32(8) (2011) 1393-1399. https://doi.org/10.3724/sp.J.1088.2011.10406.

[27] T. Xie, Z.-Y. Zhang, H.-Y. Zheng, K.-D. Xu, Z. Hu, Y.u. Lei, Enhanced photothermal catalytic performance of dry reforming of methane over Ni/mesoporous TiO2 composite catalyst, Chem. Eng. J. 429 (2022) 132507.

[28] H. Huang, M. Mao, Q. Zhang, Y. Li, J. Bai, Y. Yang, M. Zeng, X. Zhao, Solar-Light-Driven CO2 Reduction by CH4 on Silica-Cluster-Modified Ni Nanocrystals with a High Solar-to-Fuel Efficiency and Excellent Durability, Advanced Energy Materials 8 (10) (2018) 1702472, https://doi.org/10.1002/aenm.201702472.

[29] S. Wu, Y. Li, Q. Zhang, Z. Jiang, Y. Yang, J. Wu, X. Zhao, High light-to-fuel efficiency and CO2 reduction rates achieved on a unique nanocomposite of Co/Co doped Al2O3 nanosheets with UV-vis-IR irradiation, Energy Environ. Sci. 12 (8) (2019) 2581–2590, https://doi.org/10.1039/C9EE01484E.

[30] S. Wu, Y. Li, Q. Zhang, Q. Hu, J. Wu, C. Zhou, X. Zhao, Formation of NiCo Alloy Nanoparticles on Co Doped Al2O3 Leads to High Fuel Production Rate, Large Light-to-Fuel Efficiency, and Excellent Durability for Photothermocatalytic CO2 Reduction, Advanced Energy Materials 10 (42) (2020) 2002602, https://doi.org/10.1002/aenm.202002602.

[31] X. Tan, S. Wu, Y. Li, Q. Zhang, Q. Hu, J. Wu, A. Zhang, Y. Zhang, Highly Efficient Photothermocatalytic CO2 Reduction in Ni/Mg-Doped Al2O3 with High Fuel Production Rate, Large Light-to-Fuel Efficiency, and Good Durability, ENERGY &

ENVIRONMENTAL MATERIALS 5 (2) (2022) 582–591, https://doi.org/10.1002/eem2.12193.
[32] Y. Kathiraser, U. Oemar, E.T. Saw, Z. Li, S. Kawi, Kinetic and mechanistic aspects for CO2 reforming of methane over Ni based catalysts, Chem. Eng. J. 278 (2015) 62–78, https://doi.org/10.1016/j.cej.2014.11.143.
[33] S. Wu, Y. Li, Q. Hu, J. Wu, Q. Zhang, Photothermocatalytic Dry Reforming of Methane for Efficient CO2Reduction and Solar Energy Storage, ACS Sustainable Chemistry and Engineering 9 (35) (2021) 11635–11651, https://doi.org/10.1021/acssuschemeng.1c03692.
[34] M. Li, Z. Sun, Y.H. Hu, Thermo-photo coupled catalytic CO2 reforming of methane: A review, Chem. Eng. J. 428 (2022) 131222.
[35] D. Chen, L.-H. Gan, Non-catalytic direct conversion of CH4 and CO2 into high-quality syngas, Chem. Eng. J. 439 (2022) 135732.
[36] R. Buck, J.F. Muir, R.E. Hogan, Carbon dioxide reforming of methane in a solar volumetric receiver/reactor: the CAESAR project, Sol. Energ. Mater. 24 (1) (1991) 449–463, https://doi.org/10.1016/0165-1633(91)90082-V.
[37] P. Corbo, F. Migliardini, Hydrogen production by catalytic partial oxidation of methane and propane on Ni and Pt catalysts, Int. J. Hydrogen Energy 32 (1) (2007) 55–66, https://doi.org/10.1016/j.ijhydene.2006.06.032.
[38] S.Y. Foo, C.K. Cheng, T.-H. Nguyen, A.A. Adesina, Kinetic study of methane CO2 reforming on Co–Ni/Al2O3 and Ce–Co–Ni/Al2O3 catalysts, Catal. Today 164 (1) (2011) 221–226, https://doi.org/10.1016/j.cattod.2010.10.092.
[39] S.D. Angeli, G. Monteleone, A. Giaconia, A.A. Lemonidou, State-of-the-art catalysts for CH4 steam reforming at low temperature, Int. J. Hydrogen Energy 39 (5) (2014) 1979–1997, https://doi.org/10.1016/j.ijhydene.2013.12.001.
[40] V. Dal Santo, A. Gallo, A. Naldoni, M. Guidotti, R. Psaro, Bimetallic heterogeneous catalysts for hydrogen production, Catal. Today 197 (1) (2012) 190–205, https://doi.org/10.1016/j.cattod.2012.07.037.
[41] K. Takanabe, Catalytic Conversion of Methane: Carbon Dioxide Reforming and Oxidative Coupling, Journal of the Japan Petroleum Institute 55(1) (2012) 1-12. https://doi.org/10.1627/jpi.55.1.
[42] H.-Y. Zheng, Z.-Y. Zhang, K.-D. Xu, S. Wang, B.o. Yu, T. Xie, Analysis of structure-induced performance in photothermal methane dry reforming reactor with coupled optics-CFD modeling, Chem. Eng. J. 428 (2022) 131441.
[43] E. Koepf, I. Alxneit, C. Wieckert, A. Meier, A review of high temperature solar driven reactor technology: 25 years of experience in research and development at the Paul Scherrer Institute, Appl. Energy 188 (2017) 620–651, https://doi.org/10.1016/j.apenergy.2016.11.088.
[44] A. Gallo, A. Marzo, E. Fuentealba, E. Alonso, High flux solar simulators for concentrated solar thermal research: A review, Renewable and Sustainable Energy Reviews 77 (2017) 1385–1402.
[45] S.C. Rowe, A.J. Groehn, A.W. Palumbo, B.A. Chubukov, D.E. Clough, A.W. Weimer, I. Hischier, Worst-case losses from a cylindrical calorimeter for solar simulator calibration, Opt. Express 23 (19) (2015) A1309–A1323, https://doi.org/10.1364/oe.23.0a1309.
[46] T. Osório, P. Horta, M. Larcher, R. Pujol-Nadal, J. Hertel, D.W.v. Rooyen, A. Heimsath, S. Schneider, D. Benitez, A. Frein, A. Denarie, Ray-tracing software comparison for linear focusing solar collectors, AIP Conference Proceedings 1734 (1) (2016), 020017, https://doi.org/10.1063/1.4949041.
[47] Osram, xbo,, Theater Lamps. Technology and application, OSRAM GmbH, Munich, 2008.
[48] J. Costandy, N. El Ghazal, M.T. Mohamed, A. Menon, V. Shilapuram, N. Ozalp, Effect of reactor geometry on the temperature distribution of hydrogen producing solar reactors, Int. J. Hydrogen Energy 37 (21) (2012) 16581–16590.
[49] S.C. Rowe, M.A. Wallace, A. Lewandowski, R.P. Fisher, W. Ray Cravey, D. E. Clough, I. Hischier, A.W. Weimer, Experimental evidence of an observer effect in high-flux solar simulators, Sol. Energy 158 (2017) 889–897, https://doi.org/10.1016/j.solener.2017.09.040.
[50] J. Sarwar, A. Srinivasa, K. Kakosimos, Numerical investigations of the aperture size effect for maintaining a constant temperature in a novel sulfur-ammonia water splitting cycle application, Thermal Science 21 (2) (2017) 953–962, https://doi.org/10.2298/tsci141220075s.
[51] J. Sarwar, G. Georgakis, K. Kouloulias, K.E. Kakosimos, Experimental and numerical investigation of aperture size effect on the efficient solar energy harvesting for solar thermochemical applications, Energy Convers. Manage. 92 (2015) 331–341, https://doi.org/10.1016/j.enconman.2014.12.065.
[52] D. Haseler, A.M. Ali, K.E. Kakosimos, Positioning and Focusing of Light Sources in Light Concentrating Systems Using Convolutional Neural Network Modelling, Sol. Energy 218 (2021) 445–454, https://doi.org/10.1016/j.solener.2021.03.002.
[53] G. Levêque, R. Bader, W. Lipiński, S. Haussener, Experimental and numerical characterization of a new 45 kW_el multisource high-flux solar simulator, Opt. Express 24 (22) (2016) A1360–A, https://doi.org/10.1364/OE.24.0A1360.
[54] R. Shimokawa, T. Takahashi, H. Takato, A. Ozaki, Y. Takano, 2μm thin film c-Si cells on near-Lambertian Al_2O_3 substrates, Sol. Energy Mater. Sol. Cells 65 (1–4) (2001) 593–598, https://doi.org/10.1016/S0927-0248(00)00146-X.
[55] B. Bulfin, M. Miranda, A. Steinfeld, Performance Indicators for Benchmarking Solar Thermochemical Fuel Processes and Reactors, Frontiers in Energy Research 9 (2021), https://doi.org/10.3389/fenrg.2021.677980.
[56] C.W. Bale, E. Bélisle, P. Chartrand, S.A. Decterov, G. Eriksson, K. Hack, I.-H. Jung, Y.-B. Kang, J. Melançon, A.D. Pelton, C. Robelin, S. Petersen, FactSage thermochemical software and databases — recent developments, Calphad 33 (2) (2009) 295–311.
[57] C.W. Bale, P. Chartrand, S.A. Degterov, G. Eriksson, K. Hack, R. Ben Mahfoud, J. Melançon, A.D. Pelton, S. Petersen, FactSage thermochemical software and databases, Calphad 26 (2) (2002) 189–228.
[58] J. Guan, Q. Wang, X. Li, Z. Luo, K. Cen, Thermodynamic analysis of a biomass anaerobic gasification process for hydrogen production with sufficient CaO, Renew. Energ. 32 (15) (2007) 2502–2515, https://doi.org/10.1016/j.renene.2007.01.002.
[59] Y. Zhang, J. Liu, X. Lin, Z. Wang, J. Zhou, K. Cen, Detailed kinetic modeling of homogeneous HI decomposition for hydrogen production - Part II: Effect of I2 on HI decomposition, Int. J. Hydrogen Energy 38 (11) (2013) 4308–4314, https://doi.org/10.1016/j.ijhydene.2013.01.165.
[60] A.E. Kalyva, E.C. Vagia, A.G. Konstandopoulos, A.R. Srinivasa, A. T-Raissi, N. Muradov, K.E. Kakosimos, Particle model investigation for the thermochemical steps of the sulfur–ammonia water splitting cycle, International Journal of Hydrogen Energy 42 (6) (2017) 3621–3629.
[61] A.E. Kalyva, E.C. Vagia, A.G. Konstandopoulos, A.R. Srinivasa, A. T-Raissi, N. Muradov, K.E. Kakosimos, Hybrid photo-thermal sulfur-ammonia water splitting cycle: Thermodynamic analysis of the thermochemical steps, International Journal of Hydrogen Energy 42 (15) (2017) 9533–9544.
[62] Q. Pan, L. Ma, W. Du, J. Yang, R. Ao, X. Yin, S. Qing, Hydrogen-enriched syngas production by lignite chemical looping gasification with composite oxygen carriers of phosphogypsum and steel slag, Energy 241 (2022) 122927.
[63] Y. Lin, H. Wang, Z. Huang, M. Liu, G. Wei, Z. Zhao, H. Li, Y. Fang, Chemical looping gasification coupled with steam reforming of biomass using NiFe2O4: Kinetic analysis of DAEM-TI, thermodynamic simulation of OC redox, and a loop test, Chem. Eng. J. 395 (2020) 125046.
[64] P. Cao, S. Adegbite, H. Zhao, E. Lester, T. Wu, Tuning dry reforming of methane for F-T syntheses: A thermodynamic approach, Appl. Energy 227 (2018) 190–197, https://doi.org/10.1016/j.apenergy.2017.08.007.
[65] M. Jafarbegloo, A. Tarlani, A.W. Mesbah, S. Sahebdelfar, Thermodynamic analysis of carbon dioxide reforming of methane and its practical relevance, Int. J. Hydrogen Energy 40 (6) (2015) 2445–2451, https://doi.org/10.1016/j.ijhydene.2014.12.103.
[66] M.S. Challiwala, M.M. Ghouri, P. Linke, M.M. El-Halwagi, N.O. Elbashir, A combined thermo-kinetic analysis of various methane reforming technologies: Comparison with dry reforming, Journal of CO2 Utilization 17 (2017) 99–111, https://doi.org/10.1016/j.jcou.2016.11.008.
[67] A.M. Gadalla, B. Bower, The role of catalyst support on the activity of nickel for reforming methane with CO2, Chem. Eng. Sci. 43 (11) (1988) 3049–3062, https://doi.org/10.1016/0009-2509(88)80058-7.
[68] A. Vaitkus, A. Merkys, S. Grazulis, Validation of the Crystallography Open Database using the Crystallographic Information Framework, J. Appl. Crystallogr. 54 (2) (2021) 661–672, https://doi.org/10.1107/S1600576720016532.
[69] N. Kumar, S. Kanitkar, Z. Wang, D. Haynes, D. Shekhawat, J.J. Spivey, Dry reforming of methane with isotopic gas mixture over Ni-based pyrochlore catalyst, Int. J. Hydrogen Energy 44 (8) (2019) 4167–4176, https://doi.org/10.1016/j.ijhydene.2018.12.145.
[70] D.L. Trimm, Coke formation and minimisation during steam reforming reactions, Catal. Today 37 (3) (1997) 233–238, https://doi.org/10.1016/S0920-5861(97)00014-X.
[71] A. Chatla, M.M. Ghouri, O.W. El Hassan, N. Mohamed, A.V. Prakash, N.O. Elbashir, An experimental and first principles DFT investigation on the effect of Cu addition to Ni/Al2O3 catalyst for the dry reforming of methane, Appl. Catal., A 602 (2020), 117699, https://doi.org/10.1016/j.apcata.2020.117699.